

3.4 Hydroxy Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.4 Hydroxy Compounds
Difficulty	Easy

Time allowed: 30

Score: /20

Percentage: /100

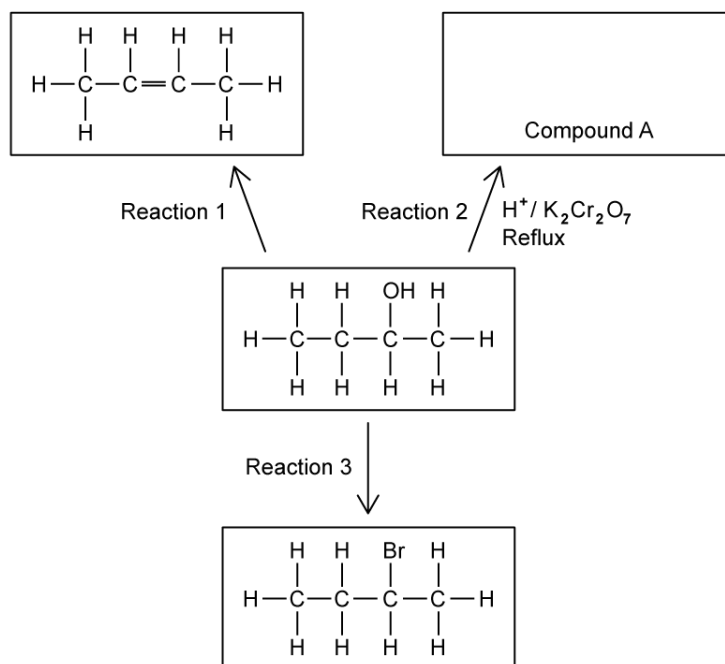
Question 1a

Butan-1-ol can undergo different reactions. State the equation for the combustion of butan-1-ol.

[2 marks]

Question 1b

A flow chart for some reactions of butan-2-ol is shown below.



State the reagents and conditions required for reactions 1 and 3.

Reaction 1

Reaction 3

[3 marks]

Question 1c

Name compound A.

[1 mark]

Question 2a

State to which class of alcohol the following alcohols belong to in Table 2.1.

Table 2.1

Alcohol	Class of Alcohol
2-methylpropan-2-ol	
Butan-1-ol	
Butan-2-ol	
2-methylpropan-1-ol	

[4 marks]

Question 2b

Write equations for

The dissociation of water

The dissociation of butan-1-ol

[2 marks]

Question 2c

Explain why alcohols are weaker acids than water.

[3 marks]

Question 3a

Three different alcohols, **A**, **B** and **C** are shown in Fig. 3.1.

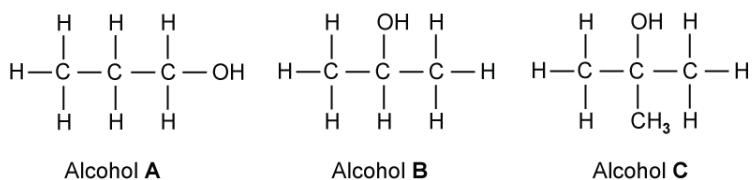


Fig. 3.1

Name alcohol **C** and state the class of alcohol to which it belongs.

Name

Class of alcohol

[2 marks]

Question 3b

Alcohol **A** is refluxed with acidified potassium dichromate.

i)

Name the organic formed

[1]

ii)

State the colour change observed in this reaction.

[1]

[2 marks]

Question 3c

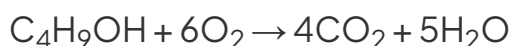
Alcohols **A**, **B** and **C** are heated separately with an alkaline solution of iodine. State which alcohol will give a yellow precipitate during this reaction.

[1 mark]

Model Answers: Easy

1a

a) The equation for the combustion of butan-1-ol is:



- Correct reactants and products; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- One common mistake when writing equations for the combustion of an alcohol is to miss out the oxygen molecule on the left hand side
- Combustion is a form of oxidation, so oxygen must be included
- To balance:
 - Balance the carbons: $\text{C}_4\text{H}_9\text{OH} + \text{O}_2 \rightarrow 4\text{CO}_2 + \text{H}_2\text{O}$
 - Balance the hydrogens: $\text{C}_4\text{H}_9\text{OH} + \text{O}_2 \rightarrow 4\text{CO}_2 + 5\text{H}_2\text{O}$
 - Balance the oxygens: $\text{C}_4\text{H}_9\text{OH} + 6\text{O}_2 \rightarrow 4\text{CO}_2 + 5\text{H}_2\text{O}$

1b

b) The reagents and conditions required for

Reaction 1:

Alternative 1

- Aluminium oxide / Al_2O_3 catalyst; [1 mark]
- Heat; [1 mark]

Alternative 2

- Concentrated sulfuric acid / phosphoric acid catalyst; [1 mark]
- Heat; [1 mark]

Reaction 3:

Alternative 1

- Sodium bromide / NaBr / potassium bromide / KBr
AND
 H_2SO_4 / H_3PO_4 (catalyst); [1 mark]

Alternative 2

- HBr
AND
Room temperature; [1 mark]

Alternative 3

- PBr_3
AND
Heat; [1 mark]

[Total: 3 marks]

- Reaction 1 is the dehydration of an alcohol – water is eliminated to form the alkene
- For reaction 3 to occur we need to have either acidic sodium or potassium bromide, or hydrogen bromide at room temperature
- This is a substitution reaction and the alcohol is transformed into a halogenoalkane

1c

c) Compound **A** is:

- Butanone; [1 mark]

[Total: 1 mark]

- Butan-2-ol is a secondary alcohol and therefore will oxidise to form a ketone
- The reaction conditions given, H^+ / $\text{K}_2\text{Cr}_2\text{O}_7$ under reflux will allow full oxidation to occur
- Butan-2-one would be accepted but the number is unnecessary as the second carbon is the only position for the $\text{C}=\text{O}$ group in butanone

2a

a) Table 2.1 should be completed as follows:

Alcohol	Class of Alcohol
2-methylpropan-2-ol	Tertiary; [1 mark]
Butan-1-ol	Primary; [1 mark]
Butan-2-ol	Secondary; [1 mark]
2-methylpropan-1-ol	Primary; [1 mark]

[Total: 4 marks]

- Alcohols can be classed as primary, secondary or tertiary
 - Primary: C-OH bonded to 1 other alkyl group (or only H)
 - Secondary: C-OH bonded to 2 other alkyl groups
 - Tertiary: C-OH bonded to 3 other alkyl groups

2b

b) The equations are:

Water

- $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{OH}^-(\text{aq}) + \text{H}^+(\text{aq});$ [1 mark]

Butan-1-ol

- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}(\text{aq}) \rightleftharpoons \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{O}^-(\text{aq}) + \text{H}^+(\text{aq});$ [1 mark]

[Total: 2 marks]

- Both water and alcohols dissociate, though only partially, so a reversible arrow must be used in the equation

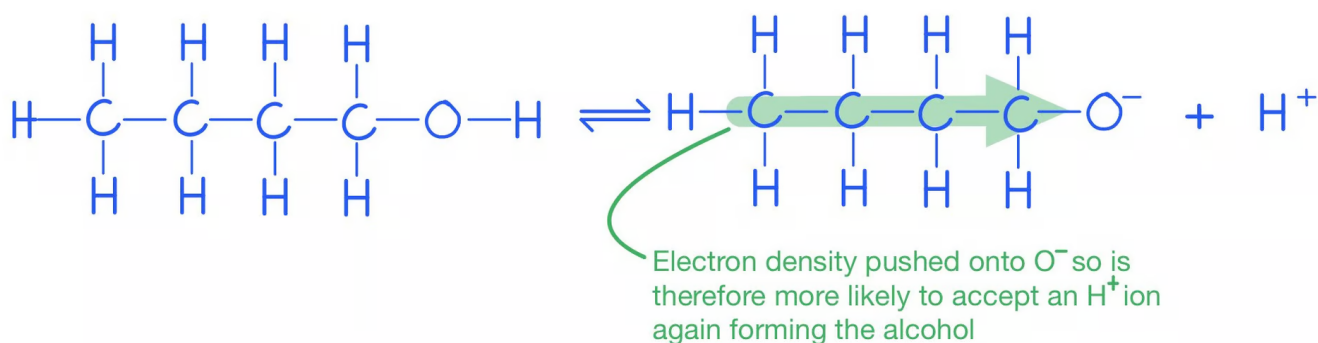
2c

c) Alcohols are weaker acids because:

- Positive inductive effect (in alcohols); [1 mark]
- More electron density on the O^- atom; [1 mark]
- More likely to accept a H^+ ion; [1 mark]

[Total: 3 marks]

- When an alcohol dissociates an alkoxide ion and hydrogen ion is formed
- Alkyl groups in the alkoxide ion donate electron density to the negatively charged oxygen, causing it to more readily accept a proton and form the alcohol again



- When water dissociates, the hydroxide ion only has one other hydrogen atom
- There is no extra electron density on the oxygen which is less likely to accept an H⁺ ion
- Water is therefore a stronger acid than alcohols

3a

a) The name and class of alcohol **C** is:

- 2-methylpropan-2-ol; [1 mark]
- Tertiary (alcohol); [1 mark]

[Total: 2 marks]

- Alcohol **C** is a tertiary alcohol
- Tertiary alcohols are alcohols in which the carbon atom bonded to the -OH group is attached to three other carbon atoms

3b

b)

i) The organic product is:

- Propanoic acid; [1 mark]

ii) The colour change is:

- Orange to green; [1 mark]

[Total: 2 marks]

- Alcohol **A** is a primary alcohol which can be oxidised to an aldehyde and then to a carboxylic acid
- If the reagents are heated under reflux the primary alcohol will oxidise full to a carboxylic acid
- The colour change of the potassium dichromate is orange to green

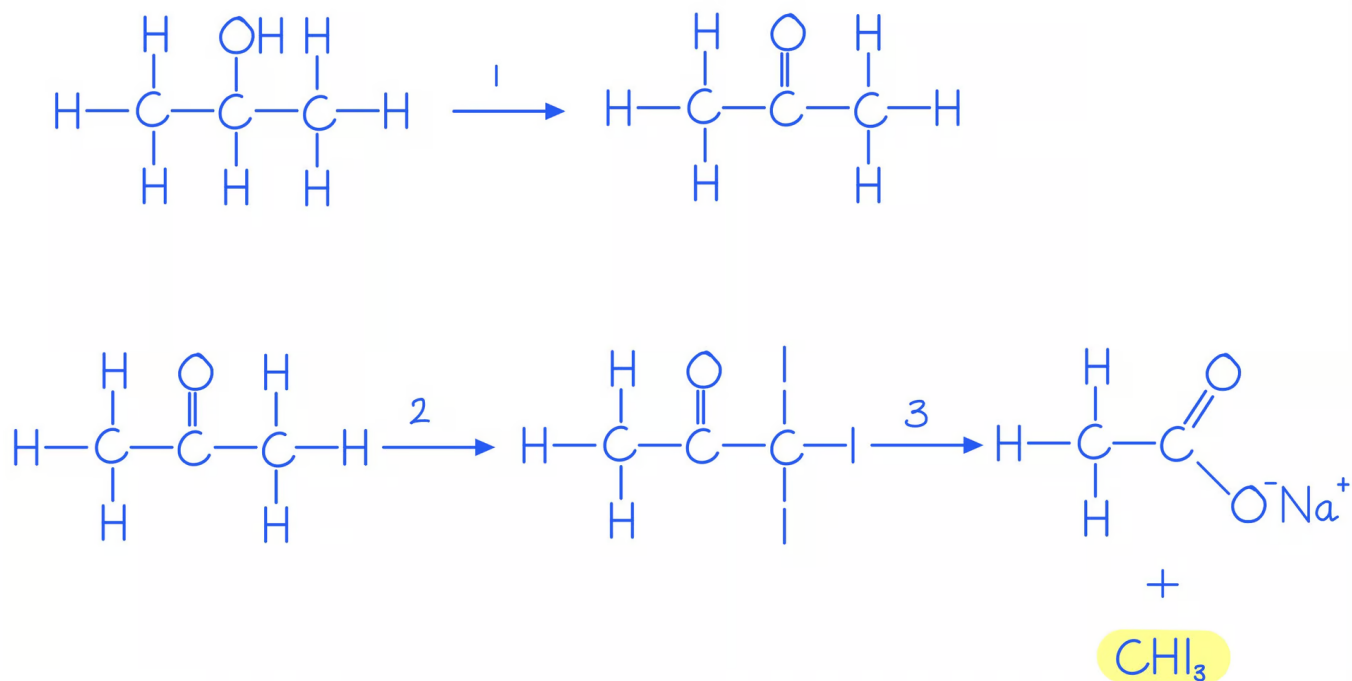
3c

c) The alcohol that will give a yellow precipitate is:

- (Alcohol) **B**; [1 mark]

[Total: 1 mark]

- Alcohol **B**, propan-2-ol, is a secondary alcohol with a methyl group bonded to the -OH carbon atom
- Firstly, in this reaction the R-OH it will get oxidised by the alkaline solution
- This will result in the formation of a methyl ketone RCOCH_3
- The methyl ketone will then first get halogenated and then hydrolysed to form the sodium salt and the yellow precipitate
 - Step 1 = oxidation
 - Step 2 = halogenation
 - Step 3 = hydrolysis

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